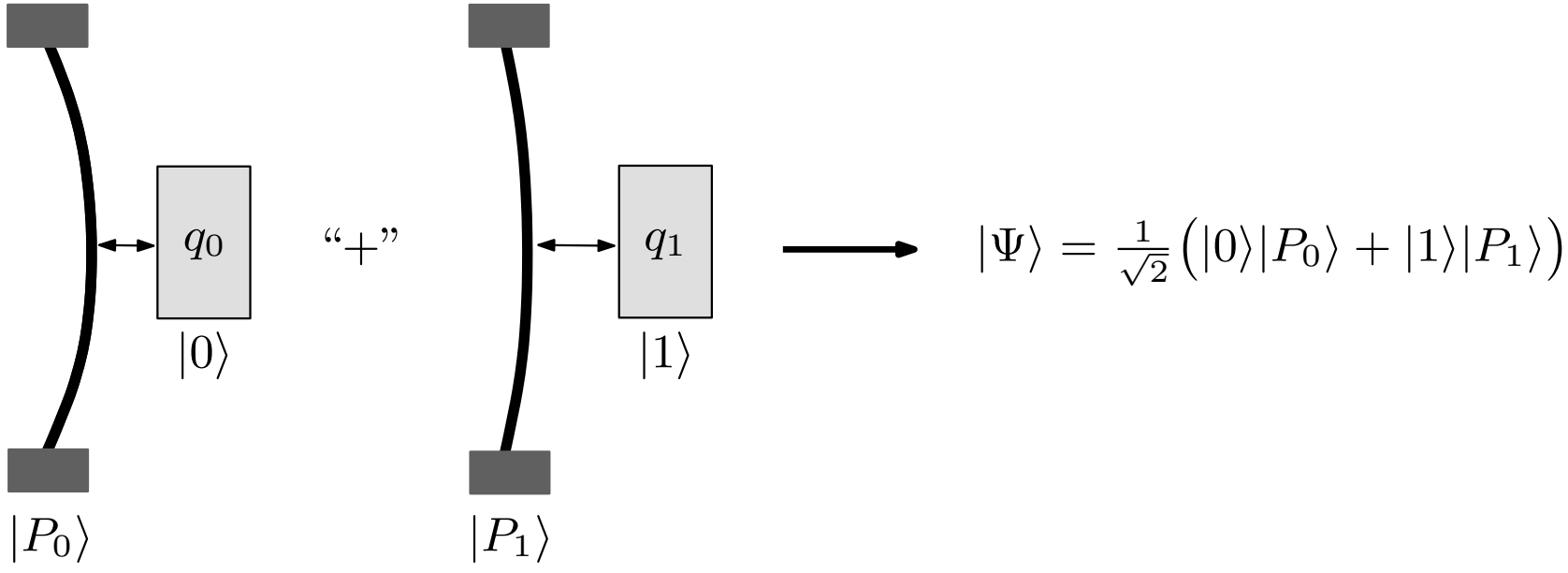

q_0
$|0\rangle$
$|P_0\rangle$
“+”
q_1
$|1\rangle$
$|P_1\rangle$
$|\Psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle|P_0\rangle + |1\rangle|P_1\rangle\big)$